

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Pharmacy SEMESTER - 8 Winter 2025 (Supply.)

Course :Bachelor of Pharmacy Branch : Pharmacy

Semester : SEMESTER - 8

Subject Code & Name: ET - 1 - BP808ET - CELL AND MOLECULAR BIOLOGY

Time : 3 Hours]

[Total Marks : 75

Instructions to the Students:

- 1.All questions are compulsory.
- 2.Draw diagrams / figures wherever necessary
- 3.Figures to right indicate full marks

Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

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|---|---|
| A) Write functions of Cell Membrane. | 2 |
| B) Define Molecular Biology. | 2 |
| C) What is the function of the Protein-Kinase Enzyme? | 2 |
| D) Define Mitosis and Meiosis. | 2 |
| E) Give significance of Protein Synthesis. | 2 |
| F) Define Cell cycle. | 2 |
| G) Elaborate Cell adaptation. | 2 |
| H) Enlist the types of RNA. | 2 |
| I) Define the term Protein & Amino acid. | 2 |
| J) What is the process of a Mutation? | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

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|---|----|
| A) Explain Misregulation of Signaling Pathways. | 10 |
| B) Describe different steps involved in Translation and Transcription process. | 10 |
| C) What is Cell Division. Explain phases of Mitosis and Meiosis with neat diagram, give the differences between them. | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

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|---|---|
| A) Differentiate between Prokaryotes and Eukaryotes. | 5 |
| B) What is the role of Second Messengers in signaling pathways? | 5 |
| C) Explain different check point in Cell cycle. | 5 |
| D) Write a note on regularities in Protein Synthesis. | 5 |
| E) Explain the Cellular Reproduction. | 5 |
| F) Write down the properties of Cell & Cell Membrane. | 5 |
| G) Write down the various DNA functioning. | 5 |
| H) Explain the structure of Protein. | 5 |
| I) What means by Meiosis? What are the four process of Meiosis? | 5 |

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